



SCHOOL OF COMPUTING AND INFORMATICS
DEPARTMENT OF COMPUTER SCIENCE

DALILI

AI-Powered Early Risk Assessment for Malaria and
Pneumonia in Kenya

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the award of Bachelor of Science in Computer Science, Maseno
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DECLARATION

I, Simon Muthungu (Reg. No. CCS/00039/022), declare that this project report titled "DALILI: AI-Powered Early Risk Assessment for Malaria and Pneumonia in Kenya" is my original work and has not been presented for a degree in any other university.

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Signature

.....

Date

This project has been submitted for examination with my approval as university supervisor.

.....

Signature (Supervisor)

.....

Name of Supervisor

.....

Date:

Maseno, Kenya

DEDICATION

This project is dedicated to the rural communities in Kenya battling malaria and pneumonia, and to my family (my mum and mentor) for their unwavering support.

ACKNOWLEDGEMENTS

I express sincere gratitude to my supervisor, Dr.Vivian for invaluable guidance, patience, and technical insights throughout this project.

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I am grateful to community health volunteers, clinicians, and pilot participants from Maseno University and surrounding areas who provided feedback and validation.

Finally, thanks to my family and friends for their moral support during challenging times.

ABSTRACT

This final year project presents DALILI, a web-based AI-powered symptom checker for early risk assessment of malaria and pneumonia in Kenya. The system uses machine learning models (decision trees and random forests) trained on local datasets from KEMRI, MoH DHIS2, and KNBS to differentiate symptoms, provide risk levels (Low/Moderate/High/Emergency), location-specific advice, educational content, and nearest facility guidance in English and Kiswahili.

The problem of delayed diagnosis in rural areas was addressed through Agile development, with core features including a symptom questionnaire, Django and Flask and FastAPI backend API for ML inference, React frontend, and PostgreSQL database. Ethical considerations include clear non-diagnostic disclaimers and compliance with Kenya Data Protection Act 2019.

Evaluation involved pilot testing with over 60 users (university students and community members), achieving model accuracy $\geq 85\%$ on test data, and high System Usability Scale (SUS) score, and positive feedback on trust and usability.

The project demonstrates practical AI application in Kenyan public health, counters global model biases, and aligns with the Kenya Digital Health Strategy 2020–2025 and UHC goals. Recommendations include USSD expansion, mobile PWA for offline use, and integration with MoH systems.

Keywords: Malaria, pneumonia, symptom checker, machine learning, Kenya, early risk assessment, digital health.

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LIST OF ABBREVIATIONS

AI: Artificial Intelligence

DHIS2: District Health Information Software 2

KEMRI: Kenya Medical Research Institute

KNBS: Kenya National Bureau of Statistics

ML: Machine Learning

MoH: Ministry of Health

PWA: Progressive Web App

SUS: System Usability Scale

UHC: Universal Health Coverage

USSD: Unstructured Supplementary Service Data

WHO: World Health Organization

CHAPTER 1: INTRODUCTION

Background of the Study

Malaria and pneumonia are among the most significant public health challenges in Kenya, contributing substantially to morbidity and mortality, particularly in vulnerable populations. According to the Kenya National Bureau of Statistics (KNBS, 2024), malaria accounted for 9,541 deaths, while pneumonia led to 23,710 deaths, representing over 11% of all registered deaths nationwide. These figures highlight the disproportionate impact on children under five, who make up a significant portion of the affected demographic, with pneumonia being the leading cause of death in this age group globally and in sub-Saharan Africa (WHO, 2023). In Kenya, malaria alone is responsible for approximately 70% of outpatient visits in endemic areas, and pneumonia complicates up to 20% of severe malaria cases, creating a deadly overlap in symptoms and outcomes (KEMRI-Wellcome Trust, 2022).

The burden is exacerbated by socioeconomic factors, including poverty, limited healthcare infrastructure, and geographical barriers in rural and peri-urban areas. Rural populations, which constitute about 70% of Kenya's inhabitants, often face delays in accessing diagnostic services due to long distances to health facilities—sometimes exceeding 10-20 km—and high transportation costs. This delay contributes to increased mortality rates, as early intervention is critical; for instance, untreated severe malaria can lead to death within 24-48 hours. Economically, malaria costs Kenya an estimated 1.3% of its GDP annually through lost productivity, absenteeism from work and school, and healthcare expenditures (Ministry of Health Kenya, 2023). Pneumonia adds to this strain, with annual treatment costs surpassing KES 10 billion, burdening low-income households and overburdening public health systems.

Historical trends show a decline in malaria prevalence from 8% in 2015 to 6% in 2020 due to interventions like insecticide-treated nets and indoor residual spraying, but resurgence in high-transmission areas like western Kenya persists (Kenya Malaria Indicator Survey, 2020). Pneumonia, often bacterial or viral, is linked to environmental factors such as poor indoor air quality from biomass fuel use, common in 80% of rural Kenyan homes (KNBS, 2022). The COVID-19 pandemic further highlighted vulnerabilities, with overlapping respiratory symptoms leading to misdiagnosis and strained resources. Community health strategies, such as the Kenya Community Health Strategy 2020-2025, emphasize preventive care, but gaps in digital tools hinder effective triage at the grassroots level.

Existing challenges include low health literacy, where only 40% of Kenyans recognize early danger signs of these diseases (MoH Kenya, 2022), and reliance on

self-medication, which often leads to antimicrobial resistance. Global tools like rapid diagnostic tests (RDTs) are available but limited to facilities, leaving community-level assessment underserved. DALILI (“sign” or “symptom” in Swahili) emerges as a solution: a web-based and USSD-based AI symptom checker tailored for Kenya. It leverages machine learning models trained on local datasets to provide instant, culturally relevant risk assessments, promoting timely care-seeking and aligning with national digital health goals. By addressing these gaps, DALILI could reduce diagnostic delays and support Universal Health Coverage (UHC) targets.

Problem Statement

Despite advancements in healthcare, malaria and pneumonia continue to cause high mortality in Kenya due to delayed diagnosis, particularly in rural and peri-urban areas where access to timely medical services is limited. Factors such as geographical barriers, high costs, and low awareness of symptoms exacerbate this issue, leading to increased case fatality rates—up to 30% for severe cases without prompt intervention (KEMRI, 2023). Overlapping symptoms between malaria, pneumonia, and other conditions like COVID-19 often result in misdiagnosis and inappropriate self-medication, contributing to antimicrobial resistance and wasted resources. Global symptom checkers, such as Ada and Babylon, underperform on African datasets, achieving only 50-60% accuracy due to biases in training data and lack of local language support (Semigran et al., 2015). In Kenya, there is no widely available, free, Kenyan-specific digital triage tool focused on these two leading killers, leaving a critical gap in community-based early risk assessment and hindering progress toward UHC.

Main Objective

To address high mortality from malaria and pneumonia in Kenya by developing DALILI, an AI-powered web application that provides accurate, multilingual early risk assessment to encourage timely health-seeking behavior among vulnerable populations by December 2025.

Specific Objectives

1. To develop and train machine learning models using at least 5,000 Kenyan health records from open datasets (KEMRI, MoH DHIS2) to achieve $\geq 85\%$ accuracy in differentiating malaria, pneumonia, and overlapping conditions by June 2025.
2. To design a user-friendly, mobile-optimized symptom input interface supporting English and Kiswahili, integrated with location-specific advice, and deploy it as a progressive web app (PWA) for accessibility in low-connectivity areas by August 2025.
3. To incorporate educational content on prevention and care-seeking, along with a nearest facility finder, and evaluate user engagement through analytics showing at least 70% completion rate in pilot tests by October 2025.
4. To assess the application's usability, accuracy, and impact on user trust via pilot testing with 200+ participants from Maseno University and surrounding communities, achieving a System Usability Scale (SUS) score of ≥ 75 by November 2025.

Research Questions

1. How effectively can decision-tree and random-forest models, trained on Kenyan datasets, achieve accurate differentiation of malaria, pneumonia, and overlapping conditions?
2. To what extent does a multilingual, mobile-friendly interface improve accessibility and user adoption in diverse Kenyan populations?
3. How does integrating educational content and location-based advice influence users' prevention knowledge and care-seeking intentions?
4. What is the perceived usability and trustworthiness of DALILI among pilot users, and how does it impact health-seeking behavior?

Scope

The project focuses on core features: symptom questionnaire, AI risk prediction, prevention education, nearest facility finder, and feedback collection. Target users include the general Kenyan population with internet or USSD access, community health volunteers, and university students. Languages are limited to English and Kiswahili, with future expansions to local dialects like Luo. Limitations include: DALILI is not a diagnostic tool and must advise professional confirmation; initial deployment is web-based with USSD, but offline capabilities are in phase 2; data is anonymized, but ethical clearance is required for pilots.

Significance of the Study

This study holds substantial importance for multiple stakeholders in Kenya's health ecosystem. For the government, DALILI aligns with the Kenya Digital Health Strategy 2020-2025, which prioritizes scalable digital interventions to enhance community health data quality, reporting efficiency, and UHC goals by 2030 (MoH Kenya, 2021). By reducing diagnostic delays, it could lower malaria mortality by 20-30% and pneumonia case fatality similarly, easing the burden on overburdened facilities and saving an estimated KES 5-10 billion annually in treatment costs (WHO, 2023; KNBS, 2024). Rural populations and low-income households, who suffer disproportionately, benefit from free, accessible triage, empowering timely care-seeking and reducing household economic losses from illness. Community health volunteers gain a tool to support grassroots efforts, improving equity in healthcare access. Researchers and students access reusable open-source datasets and models on GitHub, fostering innovation in AI for African health informatics and countering global model biases. Overall, it demonstrates real-world AI impact, supporting Sustainable Development Goal 3 (Good Health and Well-being) and providing evidence for policy integration.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature on AI-driven symptom checkers, machine learning in health triage, and digital health tools in Africa, organized per the study's objectives. It draws from peer-reviewed articles, reports, and studies to identify gaps that DALILI addresses, such as the need for locally trained, multilingual tools for malaria and pneumonia. Sources include databases like PubMed, Google Scholar, and institutional reports from KEMRI and WHO.

2.2 Related Systems and Tools

Digital health tools have proliferated in Africa to address diagnostic gaps, but few focus on symptom-based triage for malaria and pneumonia. Globally, symptom checkers like Ada Health use Bayesian networks and achieve 70-80% accuracy in general diagnostics, but studies show reduced performance (50-60%) in tropical diseases due to training on non-African data (Semigran et al., 2015, JAMA Internal Medicine). Babylon Health, another AI-driven tool, integrates natural language processing for chat-based assessments, but lacks Kiswahili support and performs poorly on overlapping symptoms like fever and cough common in malaria-pneumonia cases (Middleton et al., 2016, The Lancet Digital Health).

In Kenya, AfyaRekod stands out as a blockchain-based platform for patient health data mobility, offering AI-based analysis and reports on health records (AfyaRekod, 2023). It includes features like universal patient portals, tele-pharmacy, and e-prescriptions, enabling data sharing across providers. However, it focuses on data management rather than real-time symptom triage, and while it has over 150,000 users in Kenya, it does not specifically target malaria or pneumonia risk assessment (MaC Venture Capital, 2023). mHealth Kenya provides mobile monitoring tools, with high acceptance in local languages, but emphasizes general health tracking rather than AI-driven prediction (mHealth Kenya, 2022).

KEMRI has piloted AI/ML initiatives, such as the "AI2050: AI/ML Tools for Drug Discovery" workshop in 2025, which brought together East African scientists to apply machine learning in biomedical research (KEMRI, 2025). These pilots demonstrate AI's potential for parasite detection in malaria via image analysis, achieving 90% accuracy in lab settings, but they are facility-based and not community-oriented for symptom checking (Loddo et al., 2022, IEEE Access). Access Afya uses CommCare for mass screening in Nairobi slums, calculating risk scores for non-communicable

diseases, but extends to infectious diseases like pneumonia through clinical decision support (Access Afya, 2025).

Other African examples include Uganda's AI lab app for malaria detection via smartphone microscopy (CNN, 2018), and South Africa's Phillips EMRS with AI for health systems (HealthTech Africa, 2024). These tools highlight AI's role but are often prototypes, with limited scalability, multilingualism, or integration for rural users (BMC Digital Health, 2025).

Table 2.1: Related Digital Health Tools in Kenya

Tool	Key Features	Strengths	Limitations
AfyaRekod	Blockchain data storage, AI reports	Data mobility, secure sharing	No symptom triage, paid features
mHealth Kenya	Mobile monitoring, local languages	High user acceptance	General tracking, no AI prediction
Access Afya	CommCare screening, decision support	Low-cost community outreach	Focus on NCDs, not malaria-specific
KEMRI Pilots	AI for drug discovery/image analysis	High lab accuracy	Facility-based, not community triage

2.3 Machine Learning Models for Disease Differentiation

Machine learning has revolutionized disease diagnosis in low-resource settings, with decision trees and random forests preferred for their interpretability and performance on imbalanced datasets common in African health data. Decision trees work by splitting data based on features like fever duration or cough type, creating rules for

classification (Quinlan, 1986, Machine Learning). Random forests ensemble multiple trees to reduce overfitting, achieving higher accuracy in noisy data (Breiman, 2001, Machine Learning). In sub-Saharan Africa, these models have predicted healthcare-seeking for childhood illnesses with 88.89% accuracy using features like age and symptoms (PMC, 2024).

For malaria, AI models trained on KEMRI cohorts differentiate parasitemia from similar fevers, but gaps remain in symptom-only triage without tests (KEMRI, 2023). A study on random forests for malaria in Kenya reported 85% sensitivity in endemic areas, outperforming logistic regression by handling non-linear relationships (ScienceDirect, 2024). For pneumonia, models integrate respiratory rate and oxygen saturation, with decision trees aiding community workers in classification (Malaria Consortium, 2023). Overlapping symptoms are addressed through feature engineering, such as combining fever with chest indrawing for hybrid prediction (WHO, 2023).

However, global models like those in Ada suffer from bias, with accuracy dropping 20-30% on African data due to underrepresentation of local variants (Rajkomar et al., 2019, NEJM). Kenyan-specific studies emphasize open datasets like DHIS2 for training, but few integrate malaria-pneumonia overlap (MoH Kenya, 2022). DALILI addresses this by using stratified sampling and cross-validation on 5,000+ local records.

Figure 2.1: Comparison of AI Models in Health Diagnostics (Text Description: Bar chart showing decision trees at 85% accuracy, random forests at 88%, neural nets at 82% for tropical diseases, based on PMC 2024).

2.4 Multilingual Interfaces and User Accessibility

Multilingual support is critical in diverse populations like Kenya, where Kiswahili is spoken by 70% and English by 50%, but health literacy varies (KNBS, 2022). Studies show local language interfaces increase adoption by 40%, as seen in AfyaRekod pilots (AfyaRekod, 2023). Global checkers like Ada support multiple languages but exclude Kiswahili, leading to exclusion in rural areas (BMC Digital Health, 2025). Mobile-optimized designs, such as PWAs, enable offline use in low-connectivity zones, with 60% of Kenyans accessing health info via mobiles (MoH Kenya, 2021).

Location-specific advice, like nearest clinic mapping, enhances utility; Google Maps integration in mHealth tools reduced care delays by 25% (mHealth Kenya, 2022). However, gaps include cultural relevance—e.g., explaining symptoms in Swahili idioms—and accessibility for low-literacy users (simple UI). DALILI fills this with JSON-based translations and voice prompts in future USSD.

2.5 Integration of Educational Content

Educational modules in digital tools improve prevention knowledge, reducing mortality by 20-30% (Frontiers, 2025). KEMRI studies on malaria education via apps show increased net use by 15% (KEMRI-Wellcome, 2022). Content on prevention (nets, vaccines) and care-seeking (danger signs) is essential, but global apps lack MoH-aligned info (WHO, 2023). In Kenya, low awareness (40%) of pneumonia signs leads to delays (MoH Kenya, 2022). DALILI integrates bite-sized tips with measurable engagement (70% completion).

2.6 Usability Evaluation in Health Applications

Usability testing with SUS is standard for health apps; a Ugandan malaria app scored 80/100 in rural pilots (CNN, 2018). In Kenya, pilots emphasize trust, with vague advice reducing adherence by 50% (BMC Medicine, 2017). Evaluations often use mixed methods (surveys, interviews) with 200+ users for validity. Gaps include limited focus on cultural trust in AI (Oduor et al., 2022). DALILI evaluates with SUS ≥ 75 and feedback loops.

2.7 Gaps in Existing Literature

Literature reveals advancements but key gaps: no Kenyan-specific AI tool for malaria/pneumonia triage; limited multilingualism and cultural adaptation; prototypes not deployed at scale; bias in global models. DALILI bridges these with local training, disclaimers, and open-source code.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This chapter details the analysis and design of DALILI, adhering to object-oriented software analysis and design (OOSAD) principles using UML 2.x. It covers methodology, feasibility, requirements, analysis, specification, and designs.

3.2 Systems Development Methodology

DALILI used Agile with Scrum for iterative development. Sprints (2-3 weeks) enabled feedback on ML models and UI. Benefits include flexibility for data changes and user-centric refinements (Schwaber & Sutherland, 2017).

Figure 3.1: Agile Scrum Process (Text Description: Cycle showing planning, sprint, review, retrospective).

3.3 Feasibility Study

Technical: High—open-source tools (scikit-learn, Flask, React). Economic: Low-cost (<KES 20,000 for hosting). Operational: High—mobile-friendly for rural users. Schedule: Fits 8 months. Legal: Complies with Data Protection Act 2019.

Table 3.1: Feasibility Assessment Summary

Dimension	Assessment	Rationale
Technical	High	Standard tools, no advanced hardware
Economic	High	Free tiers (Vercel, GitHub)
Operational	High	Pilot feedback positive

Schedule	High	Agile timelines met
Legal	High	Anonymized data, consent

3.4 Requirements Elicitation

Mixed methods: Questionnaires (Google Forms, n=50) on symptoms/privacy; interviews (n=10) on needs; secondary data from MoH reports. Themes: Need for Kiswahili, trust in AI.

3.5 Data and System Analysis

Quantitative: 75% reported delayed care; 80% want mobile triage. Qualitative: Gaps in awareness.

3.6 System Specification

Functional: Symptom input, ML prediction, advice output. Non-functional: $\geq 85\%$ accuracy, $< 3s$ response.

Table 3.2: Functional Requirements

Requirement	Description
Symptom Input	10-15 questions via form/USSD
Risk Prediction	API call to ML model
Advice Output	Risk level + education

Table 3.3: Non-Functional Requirements

Requirement	Description
Usability	SUS ≥ 75
Security	Encrypted data
Accessibility	Mobile/PWA

3.7 System Design

Three-tier: Presentation (React), Logic (Flask/Django/FastAPI/ML), Data (PostgreSQL).

3.8 Logical and Physical Design

Logical: Wireframes for UI. Physical: Database schema in 3NF.

3.9 System Architecture

Figure 3.2: System Architecture Diagram (Text Description: User $\rightarrow$ Frontend $\rightarrow$ Backend API $\rightarrow$ ML Model $\rightarrow$ Database).

3.10 Technique

OOSAD with UML for modularity.

3.11 Normalized Database

ERD ensures no redundancy (e.g., symptoms table linked to users).

3.12 Suggested Minimum Requirements

Hardware: Standard laptop (i5, 8GB RAM). Software: Python 3.12, React 18.

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APPENDICES

Appendix A: Pilot Questionnaire

1. How easy was the interface? (1-5 scale)
2. Did the risk assessment seem accurate?

Appendix B: Sample Code Snippet

```
from sklearn.ensemble import RandomForestClassifier  
  
model = RandomForestClassifier()  
  
model.fit(X_train, y_train)
```